The Emergence of Mathematical Structure: A Computational Model

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1 A Computational Framework for Type-Theoretic Evolution

1.1 Preliminaries

We develop a formal framework for the incremental construction of type-theoretic structures in Cubical Type Theory (CTT). The system models the evolution of a mathematical context through discrete time steps, where new structures are selected based on their efficiency in enabling further constructions.

1.2 Basic Definitions

Definition 1 (Base Calculus)**.** We work within Cubical Type Theory (CTT) equipped with a standard *action alphabet* consisting of primitive operations:

- Type formation, introduction, and elimination rules
- Kan operations for fibrant types
- Path algebra operations (composition, transport, etc.)
- Coherence fillers for higher-dimensional structure
- Definitional equality judgments

Each primitive operation carries a *unit cost* of 1, providing a metric for computational complexity.

Definition 2 (State Space)**.** A *state* $\mathcal{B}$ is a well-typed, cumulative context in CTT, consisting of a sequence of type and term declarations. The state at time τ is denoted $\mathcal{R}(\tau)$, with initial state $\mathcal{R}(0) = \emptyset$.

Definition 3 (Effort Kernel)**.** Given a well-typed goal Y and base state $\mathcal{B}$, the *effort kernel* $K_{\mathcal{B}}(Y) \in \mathbb{N} \cup \{\infty\}$ represents the minimal number of primitive steps required to derive a sealed instance of Y from context $\mathcal{B}$. If Y is underivable from $\mathcal{B}$, we set $K_{\mathcal{B}}(Y) = \infty$.

1.3 Sealing and Integration

The notion of *sealing* ensures that new structures are properly integrated with existing ones through coherence witnesses.

Definition 4 (Context-Closed Sealing)**.** Let L_n and L_{n-1} denote the last two integration layers (see Axiom 5). We define:

(a) The *interface basis* I_n is the canonical, deduplicated set of unit schemas present in $L_n \cup L_{n-1}$, including:

- Path-lifting and transport operations
- Cubical operations (connections, reversals)

- Modality coherence schemas for structures in recent layers

(b) The *interaction profile* $J(X, \mathcal{B})$ of a candidate X at base $\mathcal{B}$ consists of all schemas from I_n that are type-theoretically applicable to X:

$$J(X, \mathcal{B}) := \{\phi \in I_n \mid \phi \text{ is applicable to } X\}$$

where applicability means that the domains and codomains of ϕ and X align such that a commutation or naturality diagram can be formed.

A candidate X is *sealed* by constructing a bundle containing:

1. The core definition of X
2. Its essential eliminators and minimal internal coherence data
3. A coherence witness (definitional equality or higher path) for each $\phi \in J(X, \mathcal{B})$

Definition 5 (Canonical Frontier)**.** Given a state $\mathcal{B}$ and effort horizon $H \in \mathbb{N}$, the *frontier basis* is the finite, deduplicated set of all canonical sealed schemas derivable within the horizon:

$$S(\mathcal{B}, H) := \{\text{sealed schemas } Y \mid K_{\mathcal{B}}(Y) \leq H\}$$

1.4 Cost and Novelty Measures

Definition 6 (Context-Closed Effort)**.** The *effort* $\kappa(X \mid \mathcal{B})$ of a sealed candidate X at base $\mathcal{B}$ combines its minimal derivation length with the cost of establishing coherence:

$$\boxed{\kappa(X \mid \mathcal{B}) := K_{\mathcal{B}}(X) + |J(X, \mathcal{B})|}$$

Definition 7 (Novelty)**.** The *novelty* $\nu(X \mid \mathcal{B}, H)$ of a sealed candidate X relative to base $\mathcal{B}$ and horizon H counts the number of schemas whose derivation becomes cheaper or newly possible:

$$\boxed{\nu(X \mid \mathcal{B}, H) := \left|\left\{Y \in S(\mathcal{B}, H) \cup S(\mathcal{B} \cup \{X\}, H) \mid K_{\mathcal{B}}(Y) - K_{\mathcal{B} \cup \{X\}}(Y) \geq 1\right\}\right|}$$

Note that if $K_{\mathcal{B}}(Y) = \infty$ and $K_{\mathcal{B} \cup \{X\}}(Y) < \infty$, the schema Y contributes 1 to the count.

Definition 8 (Efficiency and Selection Threshold)**.** The *efficiency* of a sealed candidate X is:

$$\rho(X) = \frac{\nu(X)}{\kappa(X)}$$

The Selection Threshold at step τ is given by

$$\mathrm{Bar}(\tau) = \Phi(\tau) \cdot \Omega(\tau - 1),$$

where $\mathrm{Bar}(\tau) = 0$ if $\tau = 1$, otherwise:

- $\Omega(\tau - 1) = \frac{\sum_{t<\tau} \nu(t)}{\sum_{t<\tau} \kappa(t)}$ is the historical average efficiency
- $\Phi(\tau) = \tau / \tau_{<}$, where $\tau_{<}$ is the previous realization tick.

1.5 Axioms of Evolution

The following axioms govern the dynamics of the system:

Axiom 1. Nothing Ever Disappears: Mathematics Can Only Grow.

The state evolves cumulatively while preserving well-typedness. For any time τ:

$$\mathcal{R}(\tau - 1) \subseteq \mathcal{R}(\tau)$$

Every declaration respects CTT's typing rules and universe level discipline.

Intuition: Imagine the mathematical universe as a one-way library: books can be added to the shelves, but none are ever thrown away or rewritten. Axiom 1 locks the door to erasure—every definition, once accepted, stays on the record forever. This guarantees that tomorrow's theorems can safely cite yesterday's lemmas, letting knowledge compound like interest instead of starting from scratch each morning.

Axiom 2 (Horizon Policy)**.** A global effort horizon $H \in \mathbb{N}$ governs the search space:

- **Reset:** After any successful realization, set $H \leftarrow 2$
- **Increment:** If a tick idles (no candidate selected), set $H \leftarrow H + 1$

All novelty and admissibility calculations use the current horizon H, ensuring two-step locality.

Axiom 3 (Admissibility)**.** At time n, a sealed candidate X over base $\mathcal{B} = \mathcal{R}(\tau_n - 1)$ is *admissible* if and only if:

1. X is derivable from $\mathcal{B}$ in CTT
2. Its context-closed effort satisfies $\kappa(X \mid \mathcal{B}) \leq H$

Axiom 4 (Selection Criterion)**.** From the set of admissible candidates at time τ, a candidate X is selected for realization if it clears the bar with *minimal positive overshoot*:

$$\rho(X) \geq \mathrm{Bar}(\tau) \quad \text{and} \quad X \in \underset{\substack{Y \text{ admissible} \\ \rho(Y) \geq \mathrm{Bar}(\tau)}}{\arg\min} \; (\rho(Y) - \mathrm{Bar}(\tau))$$

Tie-Breaking Protocol:

1. If multiple candidates achieve identical minimal overshoot, select those with minimal κ
2. If ties persist, realize all tied candidates in superposition, weighted by their number of distinct minimal derivations

If no admissible candidate satisfies $\rho(X) \geq \mathrm{Bar}(\tau)$, the tick idles and horizon H increments per Axiom 2.

Axiom 5 (Integration by Recomputation)**.** When candidate X_{n+1} is realized, integration work is modeled by recomputing interactions with the previous two integration layers L_n and L_{n-1}. The new *integration layer* is:

$$\boxed{L_{n+1} := (X_{n+1} \circ L_n) \uplus (X_{n+1} \circ L_{n-1})}$$

where:

- $\circ$ denotes the set of frontier schemas affected by the interaction
- $\uplus$ is disjoint union (deduplicated up to definitional equality)

The *integration gap* Δ_{n+1} represents the cost of this process, computed additively over disjoint components:

$$\Delta_{n+1} := \kappa(L_{n+1})$$

Remark 1. The framework presented here can be viewed as a computational model for the organic growth of mathematical structures, where continously increasing efficiency pressures drive the selection of new constructions.

2 Theoretical Results

This section establishes that the efficiency of realized structures grows without bound under the global selection criterion, driven by the availability of increasingly powerful abstractions.

2.0.1 The Global Bar and Variable Multiplier

Recall from Definition 8 that the selection threshold at time n is given by:

$$\mathrm{Bar}_n = \Phi(\tau_n) \cdot \Omega_{n-1}$$

where:

- $\Phi(\tau_n) = \tau_n/\tau_{n-1}$ is the *time dilation factor*
- $\Omega_{n-1} = \frac{\sum_{i=1}^{n-1} \nu_i}{\sum_{i=1}^{n-1} \kappa_i}$ is the *global average efficiency*
- By convention, $\mathrm{Bar}_1 = 0$

Axiom 4 ensures that any realized structure satisfies $\rho_n \geq \mathrm{Bar}_n$.

Lemma 1 (Amplification with Variable Multiplier)**.** *Let $K_{n-1} = \sum_{i=1}^{n-1} \kappa_i$ and $S_{n-1} = \sum_{i=1}^{n-1} \nu_i$ denote the cumulative effort and novelty through time $n-1$, with $\Omega_{n-1} = S_{n-1}/K_{n-1}$.*

If the n-th realization achieves efficiency $\rho_n \geq \Phi(\tau_n) \cdot \Omega_{n-1}$, then:

$$\Omega_n = \frac{S_{n-1} + \nu_n}{K_{n-1} + \kappa_n} \geq \Omega_{n-1} \left(1 + (\Phi(\tau_n) - 1) \cdot \frac{\kappa_n}{K_{n-1} + \kappa_n}\right)$$

Proof. Starting with the definition of Ω_n:

$$\begin{aligned}
\Omega_n &= \frac{S_{n-1} + \nu_n}{K_{n-1} + \kappa_n} = \frac{\Omega_{n-1} K_{n-1} + \rho_n \kappa_n}{K_{n-1} + \kappa_n} && (1)\\
&\geq \frac{\Omega_{n-1} K_{n-1} + \Phi(\tau_n) \Omega_{n-1} \kappa_n}{K_{n-1} + \kappa_n} && (2)\\
&= \Omega_{n-1} \left(\frac{K_{n-1} + \Phi(\tau_n)\kappa_n}{K_{n-1} + \kappa_n}\right) && (3)\\
&= \Omega_{n-1} \left(1 + (\Phi(\tau_n) - 1) \frac{\kappa_n}{K_{n-1} + \kappa_n}\right) && \square
\end{aligned}$$

2.0.2 The Fibonacci Time Dilation

The following result, which follows directly from the integration dynamics, establishes that realizations naturally occur at Fibonacci-spaced intervals.

Theorem 1 (Fibonacci Schedule and Time Dilation)**.** *Under Axiom 5 with the unit-cost model (Definition 1), the integration gaps satisfy:*

$$\Delta_{n+1} = \Delta_n + \Delta_{n-1}$$

with bootstrap conditions $\Delta_1 = \Delta_2 = 1$. Consequently:

1. *$\Delta_n = F_n$ (the n-th Fibonacci number)*
2. *$\tau_n = F_{n+1}$ for $n \geq 1$*
3. *$\Phi(\tau_n) = \frac{F_{n+1}}{F_n} \to \varphi = \frac{1+\sqrt{5}}{2}$ as $n \to \infty$*

Proof. The recurrence follows from $L_{n+1} = (X_{n+1} \circ L_n) \uplus (X_{n+1} \circ L_{n-1})$ via unit-cost invariance and additivity over disjoint union, as established in Theorem **??**. The convergence to φ follows from standard properties of consecutive Fibonacci ratios. $\square$

2.0.3 Main Result: Unbounded Growth

Theorem 2 (Unbounded Conceptual Efficiency)**.** *Let $\mathcal{R}(\tau)$ evolve according to Axioms 1–5 with the global bar (Definition 8). Under the following conditions:*

(G1) ***Fibonacci schedule:*** $\tau_n/\tau_{n-1} \to \varphi > 1$ *(Theorem 1)*

(G2) ***Minimal effort unit:*** *Every realization satisfies* $\kappa_n \geq 1$

(G3) ***Abstraction availability:*** *For any realized archetype A, there exists an admissible abstraction X with novelty $\nu(X) = \nu(A)/\kappa(X)$, yielding efficiency $\rho(X) = \nu(A)/\kappa(X)^2$*

Then the system exhibits:

1. ***Unbounded growth:*** $\limsup_{n\to\infty} \rho_n = \infty$
2. ***Acceleration via abstraction:*** *For any realized A, abstractions exist with efficiency $\rho(X) = \nu(A)/\kappa(X)^2$ that grows arbitrarily large as $\kappa(X)$ approaches its minimal description length*
3. ***Compound acceleration:*** *The global average Ω_n diverges at least polynomially in n*

Proof. Let $K_{n-1} = \sum_{i=1}^{n-1} \kappa_i$. By (G2), we have $K_{n-1} \geq n-1$, hence:

$$\frac{\kappa_n}{K_{n-1} + \kappa_n} \geq \frac{1}{n}$$

Since $\rho_n \geq \Phi(\tau_n) \cdot \Omega_{n-1}$ by selection, Lemma 1 yields:

$$\Omega_n \geq \Omega_{n-1} \left(1 + (\Phi(\tau_n) - 1) \cdot \frac{1}{n}\right)$$

By (G1), for any $\varepsilon \in (0, \varphi - 1)$, there exists N such that $\Phi(\tau_n) \geq \varphi - \varepsilon$ for all $n \geq N$. Thus:

$$\Omega_n \geq \Omega_{n-1} \left(1 + \frac{\varphi - 1 - \varepsilon}{n}\right) \quad \text{for } n \geq N$$

Iterating this inequality and using the asymptotic expansion $\prod_{k=N+1}^{n} \left(1 + \frac{\alpha}{k}\right) \sim n^{\alpha}$ yields:

$$\Omega_n \geq C \cdot n^{\varphi - 1 - \varepsilon}$$

for some constant $C > 0$. This establishes polynomial divergence of Ω_n.

For part (1), since $\rho_{n+1} \geq \Phi(\tau_{n+1}) \cdot \Omega_n$ and both factors diverge, we have $\limsup_{n\to\infty} \rho_n = \infty$.

Part (2) follows directly from (G3): for any realized archetype A with large novelty $\nu(A)$, choosing $\kappa(X)$ near its minimal value yields arbitrarily large $\rho(X) = \nu(A)/\kappa(X)^2$.

Part (3) is established by the polynomial growth of Ω_n combined with $\Phi(\tau_n) \to \varphi > 1$. □

Corollary 1 (Asymptotic φ-Boost)**.** *Under the global bar, for every $\varepsilon > 0$, there exists n_0 such that for all $n \geq n_0$:*

$$\rho_n \geq (\varphi - \varepsilon) \cdot \Omega_{n-1}$$

Proof. By selection, $\rho_n \geq \Phi(\tau_n) \cdot \Omega_{n-1}$. Since $\Phi(\tau_n) \to \varphi$ by Theorem 1, for sufficiently large n we have $\Phi(\tau_n) \geq \varphi - \varepsilon$. □

2.1 The Efficiency Dichotomy

We conclude by characterizing the fundamental difference between optimization within a fixed framework versus evolution of the framework itself.

Theorem 3 (Efficiency Dichotomy)**.** *Let $\mathcal{E}_{calc}$ denote efficiency gains achievable by calculation within a fixed formal system, and $\mathcal{E}_{evol}$ denote gains achievable through framework evolution (abstraction and language expansion). Then:*

1. ***Calculation is bounded:*** *For any two reasonable models of computation M_1, M_2, there exists a polynomial p such that:*

$$\mathcal{E}_{calc}(M_1 \to M_2) \leq p(n)$$

   This follows from the Strong Church-Turing thesis.
2. ***Evolution is unbounded:*** *For any function $f : \mathbb{N} \to \mathbb{N}$, there exists time τ such that:*

$$\mathcal{E}_{evol}(\tau) > f(\tau)$$

3. ***Orthogonal composition:*** *The total efficiency gain factors as:*

$$\mathcal{E}_{total} = \mathcal{E}_{calc} \times \mathcal{E}_{evol}$$

Proof. Part (1) is a standard consequence of computational complexity theory and the Strong Church-Turing thesis.

Part (2) follows from Theorem 2: both realized efficiencies ρ_n and the global average Ω_n diverge without bound. By condition (G3), abstractions can achieve $\rho = \nu/\kappa^2$ arbitrarily large as novel archetypes emerge.

Part (3) reflects the fundamental separation of concerns: calculation optimizes execution *within* a fixed language, while evolution optimizes the *language itself*. These orthogonal contributions compose multiplicatively. □

Remark 2. The efficiency dichotomy reveals why mathematical progress requires both computational optimization and conceptual innovation. While better algorithms provide polynomial speedups, new abstractions enable unbounded acceleration—explaining the historical pattern of mathematical breakthroughs that fundamentally restructure entire fields.

3 The Genesis Sequence

We now validate our framework by examining the sequence of mathematical structures it generates. Each realization emerges at precisely the moment when its efficiency becomes optimal, producing a progression that remarkably mirrors the historical development of mathematics.

Theorem 4 (Genesis Sequence)**.** *The first 20 structures realized by Axioms 1 - 4 are given in the table below*

τ	*Structure*	ν	κ	ν/κ	$Bar = \phi \cdot \Omega(\tau - 1)$
1	*Universe* $\mathcal{U}_0 : \mathcal{U}_1$	*1*	*2*	*0.50*	—
2	*Unit type* $\mathbf{1}$	*1*	*1*	*1.00*	*0.81*
3	*Witness* $\star : \mathbf{1}$	*2*	*1*	*2.00*	*1.62*
5	Π/Σ *types*	*5*	*3*	*1.67*	*1.62*
8	*Circle* S^1	*7*	*3*	*2.33*	*2.08*
13	*Propositional Truncation*	*8*	*3*	*2.67*	*2.59*
21	*Higher sphere* S^2	*10*	*3*	*3.33*	*2.99*
34	3-*Sphere* $S^3 \cong \mathrm{SU}(2)$	*18*	*5*	*3.60*	*3.44*
55	*Hopf fibration* $S^3 \to S^2$	*17*	*4*	*4.25*	*4.01*
89	*Generic Lie groups*	*9*	*2*	*4.50*	*4.47*
144	*Cohesion*	*19*	*4*	*4.75*	*4.67*
233	*Principal bundle connections*	*26*	*5*	*5.20*	*5.07*
377	*Curvature tensors*	*34*	*6*	*5.67*	*5.53*
610	*Metric + orthonormal frame bundle*	*43*	*7*	*6.14*	*6.05*
987	*Hilbert functional on Riemannian metrics*	*60*	*9*	*6.67*	*6.60*
1597	*Dynamical Cohesive Topos*	*150*	*8*	*18.75*	*8.29*

Proof. The automated generation of these structures based on the 4 axioms are done in Lean 4. The code is available at github.com/efficient-novelty □

The genesis sequence validates our framework in two ways: it recovers familiar mathematical milestones in a plausible order, and it discovers new structures (like the DCT) that emerge naturally from efficiency maximization. The progression from concrete to abstract, the bundling of related concepts, and the acceleration through abstraction all reflect principles that may unconsciously guide human mathematical discovery.

4 The Dynamical Cohesive Topos

At $\tau = 6765$, our framework generates a mathematical structure we call the Dynamical Cohesive Topos (DCT). While this construction builds naturally on established concepts—particularly Schreiber's cohesive $(\infty, 1)$-toposes and the well-studied theory of geometric morphisms—it represents a systematic formalization of how time actions interact with cohesive structure. The DCT synthesizes existing ideas about flows on sheaves, temporal modalities, and differential cohomology into a unified framework where the dynamical action preserves and interacts coherently with the cohesive modalities. Though similar ideas have appeared separately in various contexts, this particular combination and its axiomatic treatment provide a clean categorical framework for studying systems where geometric, logical, and temporal structures must evolve compatibly.

Throughout, let $(\mathbb{T}, \oplus, 0)$ be a (commutative) monoid object we read as "time" (e.g. $\mathbb{N}$ for discrete time or $\mathbb{R}_{\geq 0}$ for continuous time). We write $\mathrm{GeomEnd}(\mathcal{T})$ for the $(\infty, 1)$-category of geometric endomorphisms of an $(\infty, 1)$-topos $\mathcal{T}$ (i.e. inverse-image parts of geometric morphisms, with pseudonatural composition).

4.1 Definition and Axioms

Definition 9 (Dynamical Cohesive Topos (DCT))**.** A *Dynamical Cohesive Topos* is a tuple

$$\mathcal{D} \;=\; \big(\mathcal{T},\ \mathrm{coh},\ \mathsf{A}\big)$$

consisting of:

1. An $(\infty, 1)$-topos $\mathcal{T}$.

2. A *cohesive structure* over a base $(\infty,1)$-topos $\mathcal{S}$:

$$\Pi \dashv \mathrm{Disc} \dashv \Gamma \dashv \mathrm{Codisc} \ : \ \mathcal{T} \rightleftarrows \mathcal{S},$$

with the usual axioms of cohesion (equivalently, one may work with the adjoint triple $\mathrm{Shape} \dashv \flat \dashv \sharp$ relative to $\mathcal{S}$; we use either notation as convenient).

3. A *strong monoidal action* of time by geometric endomorphisms

$$\mathsf{A} \ : \ (\mathbb{T}, \oplus, 0) \longrightarrow \mathrm{GeomEnd}(\mathcal{T}), \qquad t \longmapsto \mathsf{A}_t^* \ : \ \mathcal{T} \to \mathcal{T},$$

equipped with coherent isomorphisms

$$\mathsf{A}_0^* \ \simeq \ \mathrm{id}_{\mathcal{T}}, \qquad \mathsf{A}_{s\oplus t}^* \ \simeq \ \mathsf{A}_t^* \circ \mathsf{A}_s^*,$$

and such that each A_t^* is left exact and accessible (as an inverse image functor of a geometric morphism).

These data satisfy the following *cohesion–equivariance* axioms: for each $t \in \mathbb{T}$ there are specified invertible 2-cells, natural in $X \in \mathcal{T}$,

$$\alpha_t^{\mathrm{Shape}} \ : \ \mathrm{Shape}\big(\mathsf{A}_t^* X\big) \overset{\cong}{\Longrightarrow} \mathsf{A}_t^*\big(\mathrm{Shape} X\big), \quad \alpha_t^{\flat} \ : \ \flat\big(\mathsf{A}_t^* X\big) \overset{\cong}{\Longrightarrow} \mathsf{A}_t^*\big(\flat X\big), \quad \alpha_t^{\sharp} \ : \ \sharp\big(\mathsf{A}_t^* X\big) \overset{\cong}{\Longrightarrow} \mathsf{A}_t^*\big(\sharp X\big),$$

which satisfy the usual monoidal coherence with respect to 0 and $\oplus$ and are compatible with the unit/counit data of the cohesive adjunctions (Beck–Chevalley/Frobenius conditions). If one works with the quadruple $\Pi \dashv \mathrm{Disc} \dashv \Gamma \dashv \mathrm{Codisc}$, the corresponding comparison 2-cells are required as well.

If, moreover, $\mathcal{T}$ carries *differential cohesion* (e.g. as in smooth ∞-groupoids), a DCT requires that A preserves the additional differential structure via coherent comparison 2-cells analogous to the ones above.

Remark 3 (Flows on sites). Many concrete DCT arise when a time action on a site induces pullback along "flow maps." For example, if a space X carries $a : \mathbb{T} \times X \to X$, then $\mathsf{Sh}(X)$ inherits $\mathsf{A}_t^* := a_t^{-1}$; the cohesion is the standard one relative to spaces, and the comparison maps are induced by naturality of the adjoints.

4.2 Morphisms and the category DCT

Definition 10 (Morphisms of DCTs)**.** Let $\mathcal{D}_i = (\mathcal{T}_i, \mathrm{coh}_i, \mathsf{A}^i)$ for $i = 1, 2$. A *DCT-morphism* $F : \mathcal{D}_1 \to \mathcal{D}_2$ consists of:

1. A geometric morphism $F : \mathcal{T}_1 \rightleftarrows \mathcal{T}_2$ with inverse image $F^* : \mathcal{T}_2 \to \mathcal{T}_1$ that preserves the cohesive structure (i.e. the canonical comparison 2-cells $F^*\mathrm{Shape}_2 \Rightarrow \mathrm{Shape}_1 F^*$, etc., are invertible).

2. A coherent $\mathbb{T}$-*equivariance* structure

$$\vartheta_t \ : \ F^* \circ \mathsf{A}_t^{2*} \overset{\cong}{\Longrightarrow} \mathsf{A}_t^{1*} \circ F^*$$

which is monoidal in t and compatible with the cohesion–equivariance 2-cells.

Composition is given by pasting of geometric morphisms and the monoidal natural isomorphisms. This yields:

Proposition 1 (Category and finite products)**.** *DCTs and their morphisms form an* $(\infty,1)$-*category* **DCT**. *Finite products exist: given* $\mathcal{D}_1, \mathcal{D}_2$*, the product has underlying topos* $\mathcal{T}_1 \times \mathcal{T}_2$*, product cohesion, and the componentwise action* $t \mapsto (\mathsf{A}_t^{1*}, \mathsf{A}_t^{2*})$.

4.3 Examples

(E1) Rotation on a circle. Let $X = S^1$ with $a : \mathbb{R}_{\geq 0} \times S^1 \to S^1$, $a(t,x) = x + \alpha t$ for fixed α. Set $\mathcal{T} = \mathsf{Sh}(S^1)$ with cohesion relative to spaces, and $\mathsf{A}_t^* = a_t^{-1}$. The action is strong monoidal, each A_t^* is geometric, and the cohesion–equivariance maps are induced by naturality of the adjoints.

(E2) Scaling flow on Euclidean space. Let $X = \mathbb{R}^n$ with $a(t,x) = e^{Ht}x$ $(H \geq 0)$. As above, $\mathcal{T} = \mathsf{Sh}(\mathbb{R}^n)$ and $\mathsf{A}_t^* = a_t^{-1}$ define a DCT. This models an "expanding universe" where geometric and logical structure evolve compatibly.

(E3) Discrete time via presheaves. Let $\mathcal{C}$ be a small category with an endofunctor $U : \mathcal{C} \to \mathcal{C}$. Put $\mathcal{T} = \widehat{\mathcal{C}} = \mathrm{Psh}(\mathcal{C})$, time $\mathbb{T} = (\mathbb{N}, +, 0)$, and $\mathsf{A}_n^* := (-) \circ U^n$. If $\mathcal{T}$ is given a (relative) cohesive structure, the action is left exact and the cohesion–equivariance holds whenever the adjoints are defined objectwise.

(E4) Smooth ∞-groupoids (differential cohesion). Let $\mathcal{T} = \mathsf{Sh}_\infty(\mathrm{Man})$ with differential cohesion. A smooth flow $\varphi : \mathbb{R}_{\geq 0} \times M \to M$ induces an action on $\mathsf{Sh}_\infty(M)$ by pullback $\mathsf{A}_t^* = \varphi_t^*$; the differential cohesive structure is preserved by diffeomorphisms, yielding a differential DCT.

4.4 Two basic structural facts

Proposition 2 (Invariants as a reflective subtopos; "conservation laws")**.** *Assume A_t^* preserve all small limits. The full subcategory*

$$\mathcal{T}^{\mathsf{A}} \;=\; \Big\{ X \in \mathcal{T} \;\big|\; \theta_t : \mathsf{A}_t^* X \xrightarrow{\cong} X \text{ for all } t \in \mathbb{T}, \text{ coherently} \Big\}$$

of A-invariant *objects is an $(\infty,1)$-subtopos, and the inclusion $i : \mathcal{T}^{\mathsf{A}} \hookrightarrow \mathcal{T}$ admits a left exact left adjoint (a reflector) computed as the limit of the action diagram. If the cohesion–equivariance maps are invertible, then Shape, $\flat$, and $\sharp$ restrict to $\mathcal{T}^{\mathsf{A}}$ and commute with i up to isomorphism. In particular, for any X,*

$$\mathsf{A}_t^*(\flat X) \;\cong\; \flat X, \qquad \mathsf{A}_t^*(\sharp X) \;\cong\; \sharp X,$$

i.e. the "discrete" and "codiscrete" parts are conserved.

Proof sketch. The limit of a diagram of left exact functors is left exact; invariants are the limit of the pseudodiagram $t \mapsto \mathsf{A}_t^*$. Cohesion–equivariance yields comparison 2-cells that witness commutation of modalities with the limit. □

Proposition 3 (A basic temporal modality)**.** *Suppose $\mathbb{T}$ contains a cofinal "future" submonoid $\mathbb{T}_{>0} \subseteq \mathbb{T}$, and $\mathcal{T}$ has the corresponding small limits preserved by each A_t^* and by the cohesive adjoints. Then the endofunctor*

$$\triangleright X \;:=\; \lim_{t \in \mathbb{T}_{>0}} \mathsf{A}_t^* X$$

exists and carries canonical comparison isomorphisms

$$\mathit{Shape}(\triangleright X) \;\cong\; \triangleright(\mathit{Shape} X), \qquad \flat(\triangleright X) \;\cong\; \triangleright(\flat X), \qquad \sharp(\triangleright X) \;\cong\; \triangleright(\sharp X).$$

The unit $X \to \triangleright X$ (projection to the $t > 0$ limit) serves as a "later"/guarded modality, useful for internal temporal reasoning.

Proof sketch. Limits commute with left exact functors; equivariance gives the comparison maps. The unit is induced by the inclusion of 0 into the cofinal filter of positive times. □

4.5 Remarks and open directions

- **Differential dynamics.** In a differential DCT, one can ask for conditions under which A preserves connections/curvature or sends geodesics to geodesics; Proposition 2 then identifies differential invariants.
- **Classification.** For fixed cohesive $\mathcal{T}$, classify time actions up to monoidal natural equivalence; study moduli of actions with prescribed invariants.
- **Internal type theory.** The modality $\triangleright$ of Proposition 3 underlies a temporal/guarded homotopy type theory internal to a DCT; equivariance ensures modal rules commute with Shape, $\flat, \sharp$.
- **Morphisms and limits.** Equivariant geometric morphisms form an $(\infty, 1)$-category with (finite) limits; exploring exactness properties of **DCT** is natural.

5 Analysis and Implications

This section examines why certain expected structures don't appear in the genesis sequence, explores the geometric bias of the framework, and discusses implications for understanding mathematical discovery.

5.1 Why Certain Structures Don't Appear

The absence of familiar mathematical structures like natural numbers, Boolean types, and identity types initially appears problematic. However, closer analysis reveals these structures are not missing but absorbed into more efficient realizations.

5.1.1 The Absorption Principle

The algorithm consistently favors comprehensive packages over piecemeal construction. This manifests in several ways:

Natural Numbers via Dependent Types: The natural numbers are definable as a W-type once Π and Σ types are available:

$$\mathbb{N} \equiv W_{(b:\mathbf{2})} \text{ if } b \text{ then } \mathbf{1} \text{ else } \mathbf{0}$$

Realizing $\mathbb{N}$ separately would yield efficiency approximately $\rho = 4/3 \approx 1.33$, below the bar when Π/Σ types become available. The algorithm recognizes that dependent types provide not just natural numbers but all polynomial functors, making separate realization inefficient.

Identity Types via Higher Inductive Types: The circle S^1 cannot be stated without identity types:

$$\text{loop} : \text{base} =_{S^1} \text{base}$$

Rather than realizing identity types separately, the algorithm bundles them with the first structure that genuinely requires them. The novelty $\nu(S^1) = 7$ implicitly includes the logical machinery of paths.

Set-Level Mathematics via Truncation: Boolean types, finite sets, and quotients all become expressible through propositional truncation:

$$\mathbf{2} \equiv \Sigma_{(X:\mathcal{U})} \|X\| + \|\neg X\|$$

The algorithm delays these until truncation provides a uniform framework for all set-level mathematics.

5.1.2 Efficiency Through Bundling

This absorption pattern reveals a fundamental principle: mathematical progress favors unified frameworks over isolated constructions. The efficiency gain from bundling related concepts exceeds the sum of their individual contributions:

$$\rho(\text{bundle}) > \sum_i \rho(\text{component}_i)$$

This inequality holds because bundles exhibit coherence (reducing effort) and synergy (amplifying novelty) that isolated components lack.

5.2 Geometric Bias and Physical Correspondence

The genesis sequence heavily favors geometric and topological structures over arithmetic and algebraic ones. This bias demands explanation.

5.2.1 Dimensional Advantage

The dimensional weights create an inherent efficiency advantage for higher-dimensional structures:

- 0-dimensional (arithmetic): $W(0) = 1$ per constructor
- 1-dimensional (paths): $W(1) = 5$ per constructor
- 2-dimensional (surfaces): $W(2) = 8$ per constructor
- 3-dimensional (volumes): $W(3) = 10$ per constructor

A single 1-dimensional constructor provides 5 times the novelty of a 0-dimensional one, while requiring comparable effort to define. This makes geometric structures systematically more efficient than arithmetic ones.

5.2.2 Organizational Power

Geometric structures serve as natural organizing principles for other mathematics:

- The Hopf fibration organizes the relationship between spheres
- Cohesion organizes differential geometry
- Lie groups organize continuous symmetries

Arithmetic structures, while foundational for computation, offer less organizational leverage. They excel at counting and calculation but don't naturally unify diverse mathematical phenomena.

5.2.3 Physical Correspondence

The geometric bias may reflect something deeper: physical reality itself is fundamentally geometric. The efficiency principles that drive our algorithm might mirror the principles that govern physical law:

- **Locality:** Physical laws are local, naturally expressed through differential geometry
- **Symmetry:** Conservation laws arise from continuous symmetries (Noether's theorem)
- **Dynamics:** Physical systems evolve continuously, requiring geometric rather than discrete frameworks

The algorithm's preference for geometric structures might not be a bias but a discovery: geometry provides the most efficient framework for expressing dynamical laws.

5.3 Open Problems and Future Directions

Our current framework is formulated in Cubical Type Theory, but the underlying principles may be universal across all formal systems. This leads to an ambitious conjecture:

Problem 1 (Information-Theoretic Foundations)**.** *Let U be a universal Turing machine and $K_U(X \mid Y)$ denote the conditional Kolmogorov complexity. Define mathematical evolution via:*

Effort: *$\kappa(X \mid B) = K_U(X \mid B)$ for base $B = \mathcal{R}(\tau - 1)$*

Adjacent Possible: *$\mathrm{AP}(\mathcal{R}, H)$ is the set of distinct atomic mathematical concepts (fundamental operations, rules, or templates) reachable from state $\mathcal{R}$ within effort horizon H.*

Novelty: *$\nu(X \mid \mathcal{R})$ of a candidate X relative to state $\mathcal{R}$ is the count of new atomic concepts that enter the Adjacent Possible at unit radius upon the realization of X:*

$$\nu(X \mid \mathcal{R}) = |\mathrm{AP}(\mathcal{R} \cup \{X\}, 1) \setminus \mathrm{AP}(\mathcal{R}, 1)|$$

Conjecture: *Under efficiency-driven selection with these information-theoretic definitions:*

1. *The qualitative genesis sequence (type theory $\to$ geometry $\to$ dynamics) is invariant across choice of U*
2. *Any Turing-complete formal system develops similar mathematical structures in similar order*
3. *The dimensional bias toward geometry reflects fundamental rather than contingent information-theoretic properties*

Why This Matters: If true, this would establish that mathematics is not arbitrary but determined by universal laws of information compression. Different civilizations, using different formalisms, would develop essentially the same mathematics—not because of shared psychology or physics, but because of invariant properties of algorithmic information.

Technical Challenges:

- **Uncomputability:** Kolmogorov complexity is uncomputable; proofs must work with approximations or limiting arguments
- **Invariance:** Showing results hold up to additive constants across different universal machines
- **Bridge Theorems:** Relating type-theoretic constructors to minimum description lengths
- **Empirical Validation:** Using compression algorithms to approximate K_U and test predictions

5.3.1 Implication

If either central problem is resolved—whether by proof or counterexample—it would fundamentally change our understanding of why mathematics develops as it does. We would move from describing mathematical history to understanding mathematical necessity, from observing patterns to proving laws. This transformation would have profound implications not just for mathematics, but for our understanding of knowledge, intelligence, and the nature of reality itself.

6 Mechanization in Lean 4

To validate our theoretical framework and demonstrate its computational feasibility, we have implemented the complete selection mechanism in Lean 4, a modern proof assistant and functional programming language. This implementation serves three purposes: it provides executable specifications of our axioms, it generates the Genesis sequence through pure computation rather than hand calculation, and it produces auditable certificates that can be independently verified.

6.1 Architecture of the Implementation

The implementation consists of approximately 2,500 lines of Lean 4 code organized into modular components that mirror the mathematical structure of our axioms:

CAD (Contextual Atomic Declarations): The foundation layer implements a minimal type-theoretic context discipline. Atomic declarations—universes, type formers, constructors, eliminators, computation rules, and terms—are added monotonically to contexts. The validity predicate `isValidInContext` enforces the dependency constraints that ensure well-formedness: type formers require universes, constructors require their host type, eliminators require at least one constructor (giving the correct $\nu = 2$ for early structures), and computation rules require both their eliminator and constructor to be present.

Effort Computation (Axiom 2): The `kappaMin?` function implements minimal-effort search via iterative-deepening depth-first search (IDDFS). Given a context B, target declaration Y, action alphabet, and horizon H, it returns the minimal derivation length if one exists within the bound. The search maintains monotonicity—steps that don't increase the context size are skipped to prevent loops. For efficiency, we use memoization indexed by context-state keys.

Novelty Computation (Axiom 3): The novelty module implements frontier enumeration with the schema-based keying system. The function `frontierAllScoped` computes all declarations achievable from the post-context within radius 1, then applies the cap function cap_K based on dimensional complexity. The key insight—that endogenous attachments to newly introduced type formers don't contribute to novelty—is enforced through the `excludeKeys` mechanism.

Discovery Engine (Axioms 4-5): The selection engine implements efficiency-driven discovery without pre-specified packages. The `discoverCandidates` function generates all minimal bundles X derivable under horizon H by attempting each atomic goal and extracting the delta relative to B. Foundation discipline is enforced through level checking: structures at level ℓ can only use declarations from levels ℓ and $\ell - 1$. The minimal overshoot selection rule ensures that structures emerge in order of necessity rather than jumping to premature abstractions.

6.2 Key Implementation Insights

Several design decisions were crucial for achieving correct behavior:

Sealed Bundles: When evaluating a discovered bundle X, the implementation automatically "seals" certain structures to ensure fair comparison. For instance, if X introduces a HIT-like type with constructors, we include its eliminator in the sealed version before computing effort and novelty. This prevents artificial efficiency gains from incomplete structures.

Small-Radius Frontier: Following Axiom 3, the frontier is computed at radius 1 from the post-context. This seemingly minor constraint has profound effects: it ensures that novelty measures genuine future affordances rather than distant possibilities, and it makes the computation tractable even as the context grows.

Level Stratification: The implementation maintains a level environment that assigns each declaration to its universe level. This enables the foundation check: structures can only be built using declarations from their level or below, preventing circular dependencies and ensuring mathematical coherence.

6.3 Generating the Genesis Sequence

Running `runDiscoverNTicksWithLedger` with the global action alphabet produces the Genesis sequence shown in Table 4. The implementation makes no hardcoded choices about which structures to realize—the sequence emerges purely from efficiency competition under the axioms:

```
#eval demo_rows 6765
```

This command generates the first 6765 ticks of mathematical reality, producing output that matches the theoretical predictions: the universe emerges at $\tau = 1$, the unit type at $\tau = 2$, its witness at $\tau = 3$, dependent types at $\tau = 5$, the circle at $\tau = 8$, and so on. Each realization is accompanied by a complete audit trail showing its efficiency ρ, the acceptance bar it cleared, and the frontier entries that contributed to its novelty.

6.4 Verification and Reproducibility

The implementation includes comprehensive certificate generation for independent verification. Each realized structure comes with:

- A `DerivCert` proving that the minimal derivation executes successfully
- A `FrontierEntryCert` for each novelty contribution, confirming the κ values
- A `NoveltyCert` bundling the complete novelty computation with checksums

These certificates can be validated by the `checkNoveltyCertAgainstScope` function, which re-runs the entire computation and confirms that all values match. This design ensures that the Genesis sequence is not merely computed but proven correct according to the axioms.

The complete implementation is available at `github.com/efficient-novelty`, enabling independent replication and extension of our results. By mechanizing the axioms, we demonstrate that the emergence of mathematical structure through efficiency-driven selection is not just theoretically coherent but computationally realizable.